

Customer Churn Analysis Project Guide

1. Understanding the Dataset

Target Variable:

churn → 1 = customer left, 0 = customer stayed.

Main Feature Groups:

- Customer Info: customer_id, age, gender, education, marital_status, dependents, annual_income
- Financial: monthlycharges, totalcharges, payment_method, paperless_billing, late_payments
- Services: phone service, internet service, security, backup, device protection, tech support, streaming TV/movies
- Behavior: satisfaction score, complaints, service calls, last interaction, monthly data usage
- Risk Indicators: credit_score, tenure, contract

2. Project Structure

- data/raw → original dataset
- data/processed → cleaned dataset
- notebooks → analysis and modeling notebooks
- src → reusable Python scripts
- models → saved trained models
- app → Streamlit application
- reports → EDA report, model report, business strategy
- requirements.txt → project dependencies
- README.md → project documentation

3. Notebooks Workflow

01_data_exploration.ipynb

- Check dataset shape and information
- Analyze missing values
- Visualize churn distribution
- Study relationships like churn vs tenure, complaints, payment method

02_data_preprocessing.ipynb

- Remove useless columns like customer_id
- Handle missing values
- Encode categorical variables (OneHotEncoder / LabelEncoder)
- Scale numerical features using StandardScaler

03_model_training.ipynb

- Split data into training and testing
- Train multiple models
- Logistic Regression
- Random Forest
- XGBoost

04_model_evaluation.ipynb

- Evaluate model performance
- Metrics: Accuracy, Precision, Recall, F1 Score, ROC-AUC
- Confusion Matrix visualization
- Feature Importance analysis

4. Saving the Model

Use joblib to save the trained model so it can be reused in the application.

5. Deployment (Streamlit App)

- Create a simple UI using Streamlit
- User inputs features like age, tenure, monthly charges, complaints
- The system predicts churn probability

6. Retention Strategy

- Identify high-risk customers
- If complaints > 3
- If late payments > 2
- If satisfaction score < 4
- Offer discounts, upgrades, or proactive customer support

7. Suggested Project Timeline (10 Days)

Day	Task
1	Understand dataset
2	Exploratory Data Analysis
3	Data cleaning
4	Feature engineering
5	Model training
6	Model tuning
7	Evaluation
8	Streamlit app
9	Reports
10	Presentation

8. Advanced Improvements (For Stronger CV)

- SMOTE to handle class imbalance
- SHAP to explain model predictions
- Sklearn Pipeline to automate preprocessing and modeling